

Repaso Parcial

Monday, May 18, 2020 11:11 AM

Números Complejos:

1) Sea z_1 la raíz de menor módulo de la ecuación: $z^2 + 2z + 3 + j(2z + 6) = 0$ y z_2 la de mayor módulo:
a) z_1 es imaginaria pura b) $z_2 \in \text{III cuadrante}$ c) $\bar{z}_1 = z_2$ d) z_2^2 es real e) ninguna es correcta

$$z^2 + 2z + 3 + j(2z + 6) = 0$$

$$z^2 + 2z + 3 + 2jz + 6j = 0$$

$$z^2 + (2+2j)z + 3+6j = 0$$

$$z_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$z_{1,2} = \frac{-2-2j \pm \sqrt{(2+2j)^2 - 4 \cdot 1 \cdot (3+6j)}}{2 \cdot 1}$$

$$z_{1,2} = \frac{-2-2j \pm \sqrt{4+8j-4-12-24j}}{2}$$

$$z_{1,2} = \frac{-2-2j \pm \sqrt{-12-16j}}{2}$$

$$x = \sqrt{\frac{20 + (-12)}{2}} = 2$$

$$y = \sqrt{\frac{20 - (-12)}{2}} = 4, \quad b < 0 \Rightarrow \begin{cases} + - \\ - + \end{cases}$$

$$\rightarrow z = \frac{-2 - 2j + 2 - 4j}{2} = -3j \quad (|z|=3) = \underline{z_2}$$

$$\rightarrow z = \frac{-2 - 2j - 2 + 4j}{2} = -2 + j \quad (|z|=\sqrt{5}) = \underline{z_1}$$

A) z_1 es imaginaria pura. FALSO

B) z_2 pertenece al tercer cuadrante. FALSO

C) Conjugado de $z_1 = z_2$. FALSO

D) z_2^2 es real. VERDADERO

E) Ninguna es correcta. FALSO

✓ 1) Sean w_0, w_1, w_2 y w_3 las raíces cuartas de la unidad imaginaria $z = j$, se puede asegurar que:

a) sólo una de ellas es imaginaria pura b) sólo una de ellas es real
c) $w_0 \cdot w_1 \cdot w_2 \cdot w_3 = -j$ d) $w_0 + w_1 + w_2 + w_3 = 1$ e) n.a

$$w_n = \sqrt[4]{j}$$

$$j = [1, \pi/2] = [r, \varphi]$$

$$w_n = \left[\sqrt[n]{r}, \frac{\varphi + 2 \cdot k \cdot \pi}{n} \right], \quad k = 0 \dots n-1$$

$$w_n = \left[\sqrt[n]{1}, \frac{\pi/2 + 2 \cdot k \cdot \pi}{4} \right], \quad k = 0 \dots 3$$

$$\begin{aligned} w_0 &= [1, \pi/8] \\ w_1 &= [1, 5\pi/8] \\ w_2 &= [1, 9\pi/8] \\ w_3 &= [1, 13\pi/8] \end{aligned} \quad \left. \begin{array}{l} \downarrow \pi/2 \\ \downarrow \pi/2 \\ \downarrow \pi/2 \end{array} \right\}$$

A) Solo una de ellas es imaginaria pura **FALSO**

B) Solo una de ellas es real **FALSO**

C) $w_0 \cdot w_1 \cdot w_2 \cdot w_3 = -j$

D) $w_0 + w_1 + w_2 + w_3 = 1$

E) N.A.

$$w_0 w_1 w_2 w_3 = e^{j\pi/8} e^{j5\pi/8} e^{j9\pi/8} e^{j13\pi/8}$$

$$= e^{j\frac{28\pi}{8}} = 1 \cdot e^{j\frac{7\pi}{2}}$$



Opcion ~~C~~ VERDADERA

Opcion E. **FALSA**

$$w_0 + w_1 + w_2 + w_3 =$$

$$\cos(\pi/8) + j \sin(\pi/8) + \cos(5\pi/8) + j \sin(5\pi/8)$$

$$+ \cos(9\pi/8) + j \sin(9\pi/8) + \cos(13\pi/8) + j \sin(13\pi/8)$$

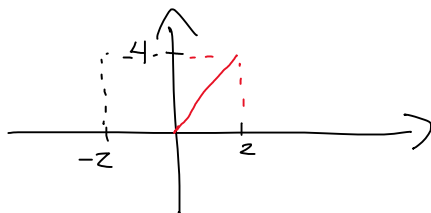
$$= 0$$

Opcion D. **FALSA**

Dada la función $f(t) = \begin{cases} 2t & \text{si } 0 < t < 2 \\ k(t) & \text{si } -2 < t < 0 \end{cases}$ y $f(t) = f(t+4)$

- a) Halle la función $k(t)$ para que la Serie Trigonómica de Fourier tenga términos con cosenos.
b) Desarrolle $f(t)$ en Serie Exponencial de Fourier.

A)



Cosenos = Par = coef. reales
(S.T.F.) (S.E.F.)

$$T = 4$$



$$\begin{cases} 2t & \text{si } 0 < t < 2 \\ -2t & \text{si } -2 < t < 0 \end{cases}$$

$$f(t) = f(t+4)$$

B) S.E.F.

$$C_m = \frac{1}{2} (a_m - b_m \cdot g)$$

$$\begin{aligned} \alpha_0 &= \frac{2}{L} \cdot \int_0^L f(t) dt \\ &= \frac{2}{2} \int_2^2 2t dt = \underline{4} \end{aligned}$$

$$\Rightarrow \frac{20}{2} = C_0 = 2$$

$$\partial_m = \frac{2}{L} \int_0^L f(t) \cos(m \cdot \omega_0 \cdot t) dt$$

$$= \frac{2}{2} \int_0^2 \underline{2t} \cdot \cos(m \cdot \pi / 2 \cdot t) dt$$

$$= 2 \cdot \left[\frac{\cos(m\pi/2t)}{(m\pi/2)^2} + \frac{t \cdot \sin(m\pi/2t)}{(m\pi/2)} \right]$$

$$= 2 \cdot \left(\frac{4(-1)^m}{m^2 \pi^2} + 0 - \frac{4}{m^2 \pi^2} - 0 \right)$$

$$\partial_n \begin{cases} n_{\text{par}} = 0 \\ n_{\text{impar}} = \frac{-16}{m^2 \pi^2} \end{cases}$$

$$C_m = \frac{1}{2} \partial m = - \frac{B}{m^2 \pi^2}$$

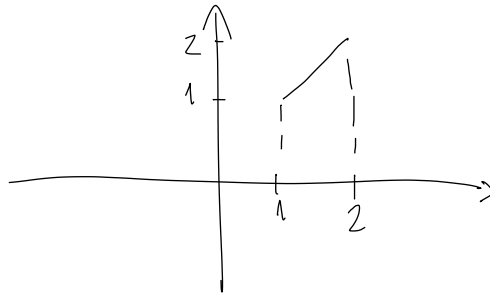
$$S_{E.F.} = S(t) = 2 - \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n+1)^2} e^{-(2n+1)\pi/2 \cdot t}$$

$$= 2 - \frac{8}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cdot \left(e^{j(2k+1)\pi/2 \cdot t} + e^{-j(2k+1)\pi/2 \cdot t} \right)$$

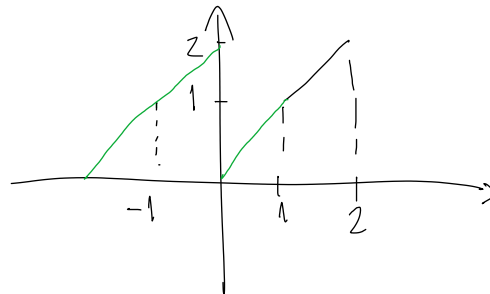
Ejercicio n° 2: a) Dada $f(t) = t$ en $(1,2)$, complete en $(0,1)$ con $T=2$ para que la STF no tenga cosenos, y su valor medio sea 1. Desarrolle la serie.

b) A partir de la serie hallada, calcule el valor de $\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}$

A)



Me piden impar desplazada (V.M.=1)



$$f(t) = t \text{ si } 0 \leq t \leq 2$$

$$f(t) = f(t+2)$$

$$T=2$$

$$L=1$$

$$\omega_0 = \pi$$

$$\frac{\partial \phi}{\partial z} = 1$$

$$\partial_m = 0$$

$$b_m = \frac{1}{1} \cdot \int_0^2 t \cdot \sin(m\pi t) dt = \left[\frac{\sin(m\pi t)}{m^2 \pi^2} - \frac{t \cdot \cos(m\pi t)}{m\pi} \right]_0^2$$

$$= \frac{-2}{m\pi}$$

$$S(t) = 1 - \frac{2}{\pi} \sum_{m=1}^{\infty} \frac{1}{m} \sin(m\pi t)$$

B) Si $t = \frac{1}{2}$

m	$\sin(m\pi/2)$
1	1
2	0
3	-1
4	0
5	1

Sólo armónicas impares

$$\begin{matrix} 0 & | & 0 \\ \vdots & & \vdots \end{matrix} \downarrow$$

$$f\left(\frac{1}{2}\right) = \frac{1}{2} = 1 - \frac{2}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}$$

$$\frac{2}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{1}{2}$$

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}$$

Ej. 2: Dada la siguiente ecuación diferencial:
 $y' - y = 2t - t^2$ con $y(0)=1$
 a) Resuelva analíticamente por Transformada de Laplace.

$$y'(t) - y(t) = 2t - t^2, \quad y(0)=1$$

Laplace m.a.m.

$$\mathcal{L}\{y'(t) - y(t)\} = \mathcal{L}\{2t - t^2\}$$

$$Y(s)(s-1) = \frac{2}{s^2} - \frac{2}{s^3} + 1$$

$$Y(s) = \frac{2s - 2 + s^3}{s^3(s-1)}$$

$$\rightarrow \frac{s^3 + 2s - 2}{s^3(s-1)} = \frac{\overset{-2}{-1} \cdot 2}{s^3} + \frac{B}{s^2} + \frac{C}{s} + \frac{\overset{1}{1} \cdot 1}{s-1}$$

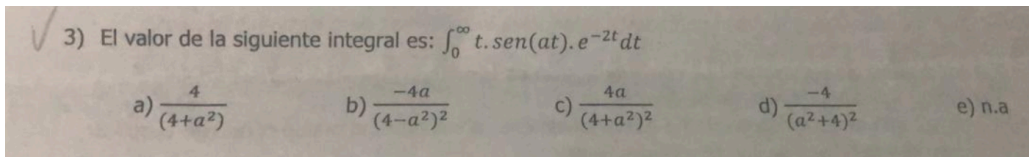
$$2(s-1) + B s(s-1) + C s^2(s-1) + 1 s^3 = s^3 + 2s - 2$$

Cúbico: $C + 1 = 1 \Rightarrow C = 0$

Cuadrático: $B = 0$

$$Y(s) = \frac{2}{s^3} + \frac{1}{s-1}$$

$$\mathcal{L}^{-1}(Y(s)) = y(t) = t^2 + e^t$$



Def. Transf. Laplace:

$$\mathcal{L}(f(t)) = F(s) = \int_0^{\infty} \underline{f(t)} \underline{e^{-st}} dt$$

$$\int_0^{\infty} \underline{t \cdot \text{sen}(at)} \underline{e^{-2t}} dt$$

$$= \mathcal{L}[t \cdot \text{sen}(at)]_{[2]} \checkmark$$

Mult. $\times t^m$. ($m=1$)

$$1^{\circ}) \text{ Derivo) } \frac{d}{ds} \left(\frac{a}{s^2 + a^2} \right)$$

$$= \frac{-2 \cdot a \cdot s}{(s^2 + a^2)^2}$$

2^o) Cambio signo (una vez)

$$\mathcal{L}[t \cdot \text{sen}(at)] = \frac{2 \cdot a \cdot s}{(s^2 + a^2)^2}$$

$$\mathcal{L}[t \cdot \text{sen}(at)]_{[2]} = \frac{4a}{(4 + a^2)^2}$$

(opción C)

Sea $G(s) = \frac{s(s^2 - 4s + 12)}{(s^2 + 4s + 5)(s^2 - 3s)}$ la transferencia de un sistema.

a) Halle el valor de $K \in \mathbb{R}$ sabiendo que el sistema es estable, y grafique aproximadamente el corte de $|G(s)|$ con el eje real.

b) Halle la respuesta del sistema a una entrada impulso unitario

$$A) \quad G(z) = \frac{25(z^2 - kz + 12)}{(z^2 + 4z + 5)(z^2 - 3z)}$$

$$= \frac{25(z^2 - kz + 12)}{\underbrace{(z^2 + 4z + 5)}_{\text{Comp con } \sqrt{\text{parte real}} < 0} \cdot \underbrace{z(z - 3)}_{\text{Real Positiva}}}$$

[Marg. estable]
es estable]

$$[z^2 - kz + 12]_{[3]} = 0$$

$$9 - 3k + 12 = 0$$

$$\boxed{k = 7}$$

$$\begin{array}{r|rrr} 3 & 1 & -7 & 12 \\ & & 3 & -12 \\ \hline & 1 & -4 & 0 \end{array}$$

$$G(z) = \frac{25(z - 4)}{(z^2 + 4z + 5) \cdot z}$$

$$z_1 = 4$$

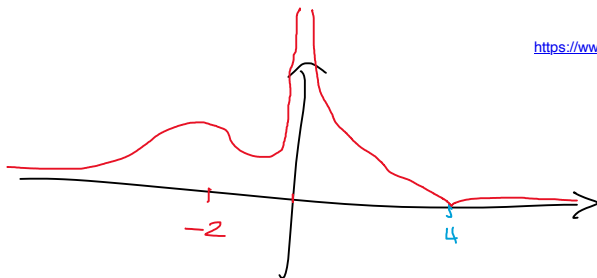
$$z_2 = \infty$$

$$p_1 = -2 + j$$

$$p_2 = -2 - j$$

$$p_3 = 0$$

Corte eje real.



<https://www.geogebra.org/classic/pqluvdtj>

$$B) \quad R_{t2} \approx \underline{\underline{y(t)}}$$

$$G(z) F(z) = Y(z)$$

$$- \dots \underbrace{\dots}_{\downarrow} \dots$$

$$G(z) = Y(z)$$

$$Y(z) = \frac{25(z-4)}{z(z^2+4z+5)}$$

$$= \frac{A}{z} + \frac{Bz+C}{z^2+4z+5}$$

$\frac{-100}{5} = -20$

$$-20(z^2+4z+5) + Bz^2 + Cz = 25z - 100$$

Cuadráticos: $-20 + B = 0 \Rightarrow B = 20$

Lineales: $-80 + C = 25 \Rightarrow C = 105$

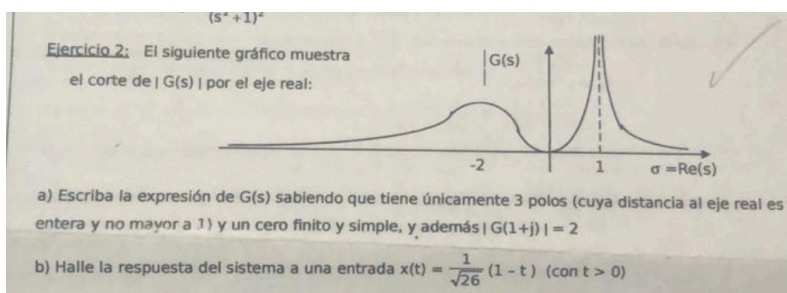
$$Y(z) = \frac{-20}{z} + \frac{20z+105}{z^2+4z+5}$$

$$Y(z) = \frac{-20}{z} + \frac{20z+105}{(z+2)^2+1}$$

$40+65$

$$Y(z) = \frac{-20}{z} + \frac{20 \cdot (z+2)}{(z+2)^2+1} + \frac{65}{(z+2)^2+1}$$

$$y(t) = -20 + 20 \cos(t) \cdot e^{-2t} + 65 \sin(t) e^{-2t}$$

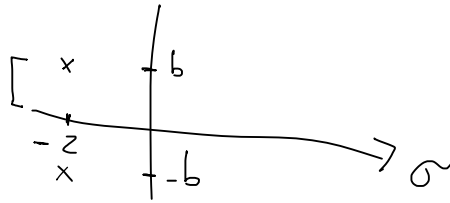


$$Z_1 = 0$$

$$P_1 = 1$$

$$\left[\begin{array}{l} P_2 = -2 + b\gamma \\ P_3 = -2 - b\gamma \end{array} \right.$$

$$|G(1+\gamma)| = 2$$



b es entero, no mayor que 1
 y las raíces son conjugadas

$$\boxed{b=1}$$

Su polinomio es

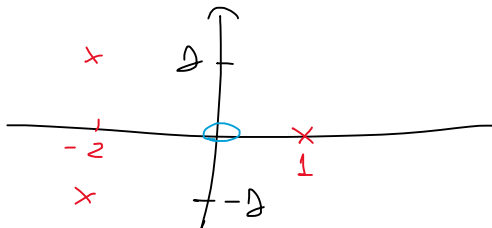
$$[z - (-2 + \gamma)] [z - (-2 - \gamma)]$$

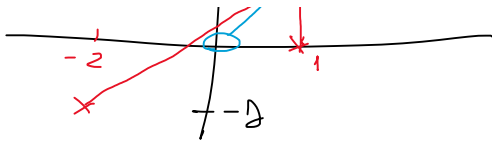
$$= z^2 + 2z + 2\gamma + 2z + 4 - 2\gamma - \gamma z + 2\gamma + 1$$

$$= z^2 + 4z + 5$$

$$G(z) = \frac{k \cdot z}{(z-1)(z^2+4z+5)}$$

Constelación. $|G(1+\gamma)| = 2$





$$|G(1+j)| = \frac{k \sqrt{2}}{3 \cdot \sqrt{3} \cdot 1} = 2$$

$$k = 3 \cdot \sqrt{26}$$

$$G(s) = \frac{3\sqrt{26} \cdot s}{(s-1)(s^2+4s+5)}$$

B) rta. a entrada

$$x(t) = \frac{1}{\sqrt{26}} (1-t), \quad t > 0$$

$$Y(s) = G(s) \cdot X(s)$$

$$X(s) = \frac{1}{\sqrt{26}} \left(\frac{1}{s} - \frac{1}{s^2} \right) = \frac{1}{\sqrt{26}} \left(\frac{s-1}{s^2} \right)$$

$$Y(s) = \frac{3 \cdot \sqrt{26} \cdot s}{(s-1)(s^2+4s+5)} \cdot \frac{s-1}{s^2} \cdot \frac{1}{\sqrt{26}}$$

$$Y(s) = \frac{3}{s(s^2+4s+5)}$$

$$Y(s) = \frac{\frac{3}{5}}{s} + \frac{B s + C}{s^2+4s+5}$$

$$\frac{3}{5} (s^2+4s+5) + B s^2 + C s = 3$$

$$\text{Comparativos. } \frac{3}{5} + B = 0 \Rightarrow B = -\frac{3}{5}$$

Lineales: $\frac{12}{5} + C = 0 \Rightarrow C = -\frac{12}{5}$

$$\rightarrow Y(z) = \frac{(3/5)}{2} - \frac{\overbrace{(3/5)z + 12/5}^{\frac{6}{5} + \frac{6}{5}}}{(z+2)^2 + 1}$$

$$Y(z) = \frac{(3/5)}{2} - \frac{3}{5} \frac{(z+2)}{(z+2)^2 + 1} - \frac{6/5}{(z+2)^2 + 1}$$

$$y(t) = \frac{3}{5} - \frac{3}{5} \cos(t) e^{-2t} - \frac{6}{5} \sin(t) e^{-2t}$$

✓ Ejercicio nº 4:
Usando Transformada Z hallar la solución de la siguiente ecuación en diferencia:
 $x(n+1) = 3x(n) + 2 \quad x(0) = 2$
Verifique el resultado obtenido hallando el valor $x(2)$ usando la ecuación en diferencia y su solución

$$X(n+1) = 3X(n) + 2, \quad \underline{X(0) = 2}$$

Transf Z m.a.m.

$$z \cdot X(z) - z \cdot \underline{X(0)} = 3X(z) + \frac{2z}{z-1}$$

$$X(z)(z-3) = \frac{2z}{z-1} + 2z$$

$$X(z) = \frac{2z^2}{(z-1)(z-3)}$$

Frac. Simples:

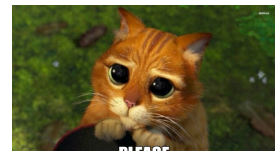
DEJAR UNA Z AFUERA!!!!!!

$$\frac{2z}{-z} = -1$$

↓
1

$$\frac{6}{2} = 3$$

↓
0



$$\frac{-}{(z-1)(z-3)} = \frac{A}{z-1} + \frac{B}{z-3}$$

$$X(z) = -\frac{z}{z-1} + \frac{3z}{z-3}$$

$$X(m) = -1 + 3(3)^m = -1 + 3^{m+1}$$

Con la ec. en diferencia:

$$X(0) = 2$$

$$x(n+1) = 3x(n) + 2 \quad x(0) = 2$$

$$X(1) = 3 \cdot X(0) + 2 = 8$$

$$X(2) = 3 \cdot X(1) + 2 = 26 \quad \checkmark$$

Con la $X(m)$ hallada.

$$X(2) = -1 + 3 \cdot 3^2 = 26 \quad \checkmark$$

5) La región de convergencia de transformada Z de $x(n) = \begin{cases} -3 & \text{si } n \text{ no múltiplo de } 3 \\ (-2)^n & \text{si } n \text{ es múltiplo de } 3 \end{cases}$

a) $|z| > 2$ b) $|z| \geq 3$ c) $|z| \geq 2$ d) $|z| < 3$ e) $2 < |z| < 3$ f) n.a

$$X(z) = \sum_{n=0}^{\infty} X(n) \cdot z^{-n}$$

$$= \frac{(-2)^0}{z^0} + \frac{-3}{z^1} + \frac{-3}{z^2} + \frac{(-2)^3}{z^3} + \frac{-3}{z^4} + \dots$$

$$= \sum_{k=0}^{\infty} \left(\frac{-2}{z} \right)^{3k} - 3 \cdot \sum_{k=0}^{\infty} \left(\frac{1}{z} \right)^{3k+1} - 3 \cdot \sum_{k=0}^{\infty} \left(\frac{1}{z} \right)^{3k+2}$$

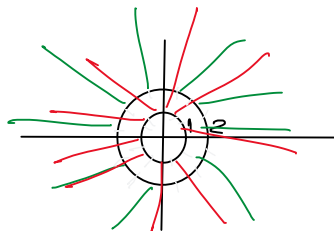
$$= \sum_{k=0}^{\infty} \left(\frac{-8}{z^3} \right)^k - \frac{3}{z} \sum_{k=0}^{\infty} \left(\frac{1}{z^3} \right)^k - \frac{3}{z^2} \sum_{k=0}^{\infty} \left(\frac{1}{z^3} \right)^k$$

Sólo importa la región de convergencia:

$$\sum_{k=0}^{\infty} z^k \text{ Converge si, y sólo si, } |z| < 1 \quad \left(\text{Converge a } \frac{1}{1-z} \right)$$

$$\left| \frac{-8}{z^3} \right| < 1 \Rightarrow |z^3| > 8$$
$$|z| > 2$$

$$\left| \frac{1}{z^3} \right| < 1 \Rightarrow |z^3| > 1$$
$$|z| > 1$$



Intersección
 $|z| > 2$

Para practicar, vamos a hallar la transformada

$$X(z) = \sum_{k=0}^{\infty} \left(\frac{-8}{z^3} \right)^k - \frac{3}{z} \sum_{k=0}^{\infty} \left(\frac{1}{z^3} \right)^k - \frac{3}{z^2} \sum_{k=0}^{\infty} \left(\frac{1}{z^3} \right)^k$$

$$= \frac{1}{1 - \left(\frac{-8}{z^3} \right)} - \frac{3}{z} \cdot \frac{1}{1 - \left(\frac{1}{z^3} \right)} - \frac{3}{z^2} \cdot \frac{1}{1 - \left(\frac{1}{z^3} \right)}$$

$$= \frac{z^3}{z^3 + 8} - \frac{3}{\cancel{z}} \cdot \frac{z^{\cancel{3}2}}{z^3 - 1} - \frac{3}{\cancel{z^2}} \cdot \frac{z^{\cancel{3}1}}{z^3 - 1}$$

$$= \boxed{\frac{z^3}{z^3 + 8} - \frac{3z^2}{z^3 - 1} - \frac{3z}{z^3 - 1}}$$

$$\boxed{z^3 + 8 \quad z^3 - 1 \quad z^3 - 1}$$

Si sobra tiempo:

Ejercicio nº 1: a) Halle el valor de $n \in \mathbb{N} / (1 - 2j)^n = 117 - 44j$ (Justifique su forma de hallarlo)

b) Resuelva en $\mathbb{C}$: $z^4 + 16 = 0$ $\cap$ $|z + 1 - j| \leq 1$

$$\begin{aligned} A) \quad (1 - 2j)^n &= [\sqrt{5}, \arctg(-2)]^n = [(\sqrt{5})^n, n \cdot \arctg(-2)] \\ &= 117 - 44j = [125, \arctg\left(\frac{-44}{117}\right)] \end{aligned}$$

$$(\sqrt{5})^n = 125$$

$$n = 6 \quad \checkmark$$

Chequear los argumentos (llevar todo a 1º giro positivo)

$$\arctg\left(\frac{-44}{117}\right) = -0,3597 + 2\pi = 5,9235$$

$$6 \cdot \arctg(-2) = 6 \cdot (-1,1072 + 2\pi) = 31,0562$$

$$\begin{aligned} &\downarrow \\ &\underline{-8\pi \text{ (4 giros)}} \\ &5,9235 \quad \checkmark \end{aligned}$$

$$B) \quad \textcircled{1} \quad z^4 + 16 = 0 \quad \wedge \quad \textcircled{2} \quad |z + 1 - j| \leq 1$$

① En polar:

$$z^4 = -16$$

$$Z = \sqrt[4]{[16, \pi]}$$

$$\omega_0 = [2, \pi/4] = \sqrt{2} + \sqrt{2}j$$

$$\omega_1 = [2, 5\pi/4] = -\sqrt{2} + \sqrt{2}j$$

$$\omega_2 = [2, 9\pi/4] = -\sqrt{2} - \sqrt{2}j$$

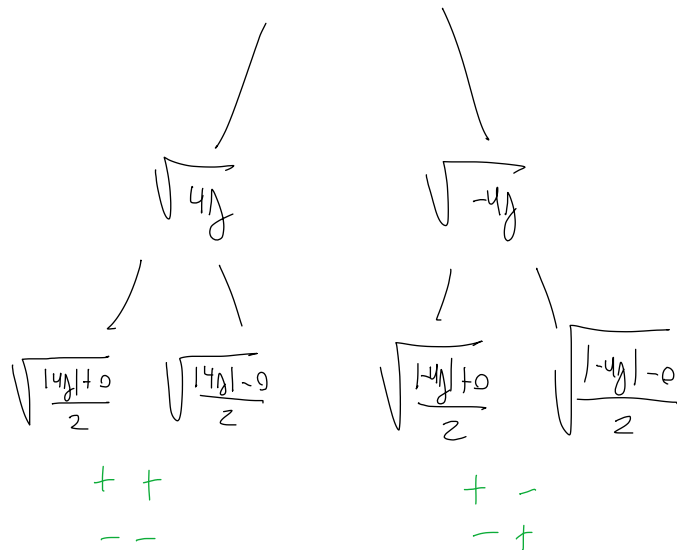
$$\omega_3 = [2, 13\pi/4] = \sqrt{2} - \sqrt{2}j$$

En binómica:

$$Z^4 = -16$$

$$Z = \sqrt[4]{-16}$$

$$Z = \sqrt{\sqrt{-16}}$$



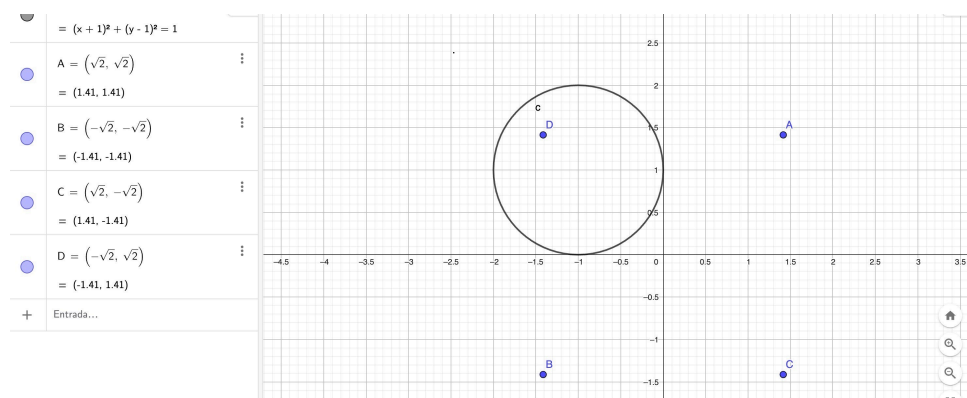
$$\omega_0 = \sqrt{2} + \sqrt{2}j$$

$$\omega_1 = -\sqrt{2} - \sqrt{2}j$$

$$\omega_2 = \sqrt{2} - \sqrt{2}j$$

$$\omega_3 = -\sqrt{2} + \sqrt{2}j$$

$$(2) \quad |Z + 1 - j| \leq 1$$



Solo cumple $z = -\sqrt{2} + \sqrt{2}y$

Sino :

$$|\sqrt{2} + \sqrt{2}y + 1 - y| = \sqrt{6} \approx 2,4495 \quad \times$$

$$|-\sqrt{2} - \sqrt{2}y + 1 - y| = \sqrt{6} \quad \times$$

$$|\sqrt{2} - \sqrt{2}y + 1 - y| = 2 + \sqrt{2} \approx 3,4142 \quad \times$$

$$|-\sqrt{2} + \sqrt{2}y + 1 - y| = 2 - \sqrt{2} \approx 0,5858 \quad \checkmark$$